UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Scalar and Predictive Quantization

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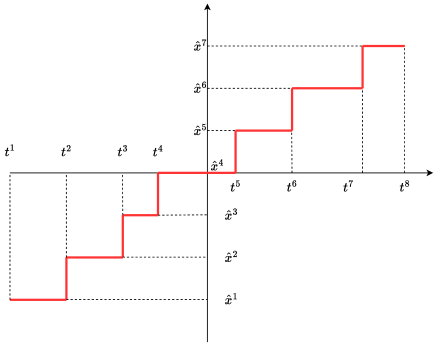
Multimedia Communications

Outline

Outline

Scalar quantization (SQ): it is just a staircase function (almost always)



Here we show the graph of a possible $Q(x)$

Go to **wooclap.com** with the code **PBTFUO**

Input range:

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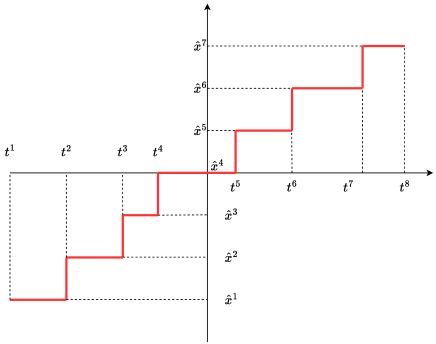

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Input range: (t_1, t_8)

Number of levels:

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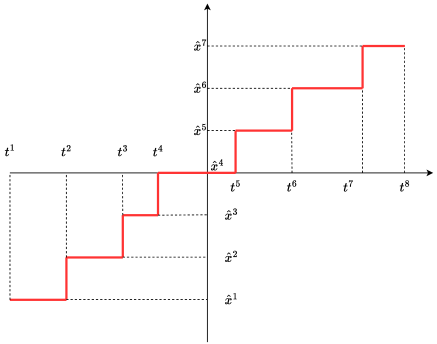

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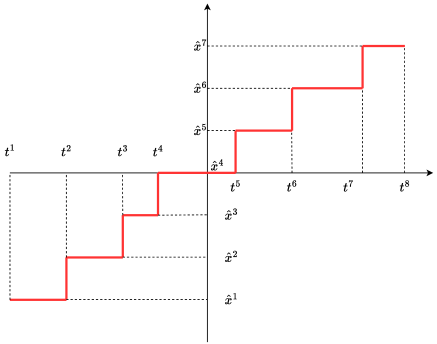
Input range: (t_1, t_8)

Number of levels: $L = 7$

Scalar quantization (SQ): it is just a staircase function (almost always)



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Input range: (t_1, t_8)

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The t^i's determine a *partition* of the input interval

Each $\Theta^i = \left(t^i, t^{i+1}\right)$ is a quantization region or cell

Any number in $\Theta^i = \left(t^i, t^{i+1}\right)$ is quantized as $\hat{x}^i$

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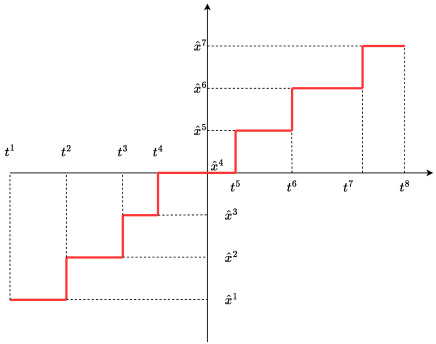

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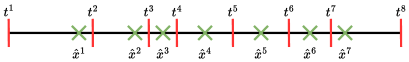
The graph below also represent the quantizer: the thresholds and the levels competely determine the quantizer



Definition : scalar quantization (SQ)

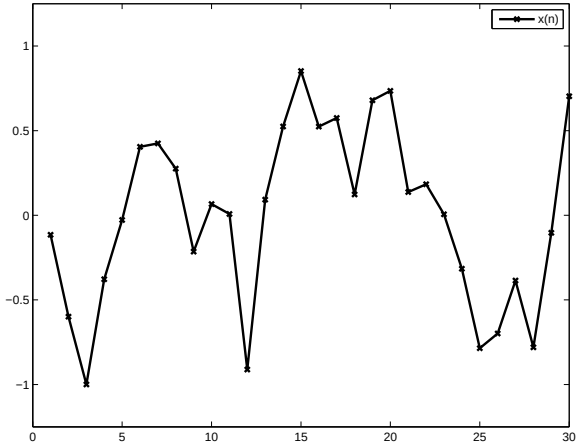
Scalar quantization (SQ) is a function Q from $\mathbb{R}$ to a discrete set called *Dictionary*

$$Q : x \in \mathbb{R} \rightarrow y \in \mathcal{C} = \{\hat{x}^1, \hat{x}^2, \dots \hat{x}^L\} \subset \mathbb{R}$$

- C : Dictionary, it is a discrete subset of $\mathbb{R}$
- $\hat{x}^i$: quantization level, code-word
- $e = x - Q(x)$: Quantization noise
- $\Theta^i = \left\{x : Q(x) = \hat{x}^i\right\}$: Decision regions or cells

Regions and levels completely define the SQ
Regions are typically defined as intervals: $\Theta^i = \left(t^i, t^{i+1}\right)$
Therefore, thresholds and levels also completely define a SQ

Example 1



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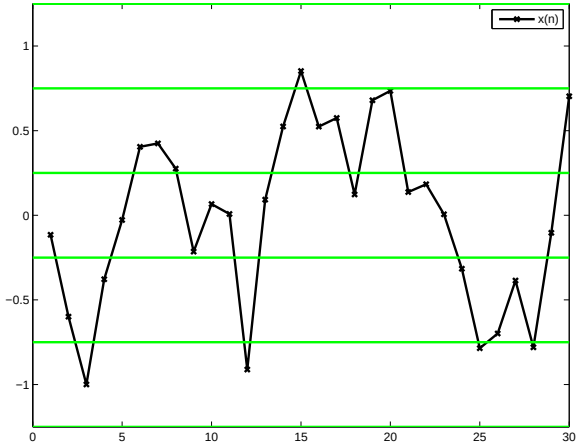

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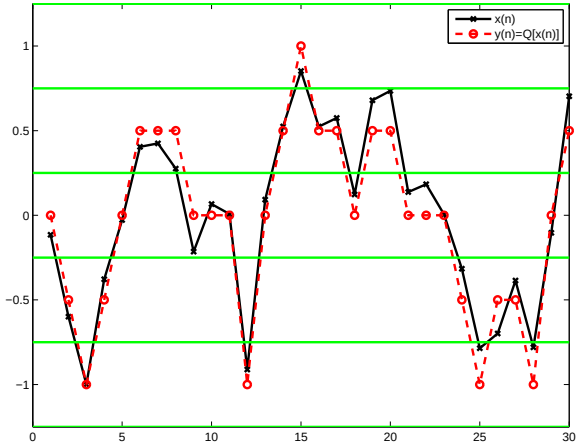

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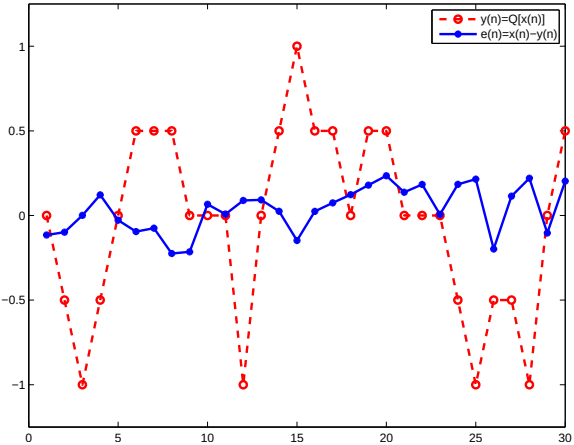

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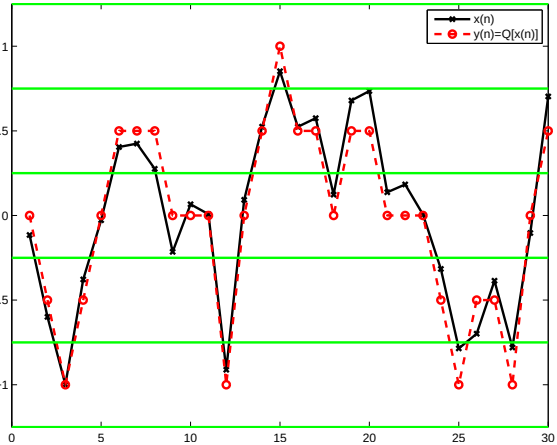
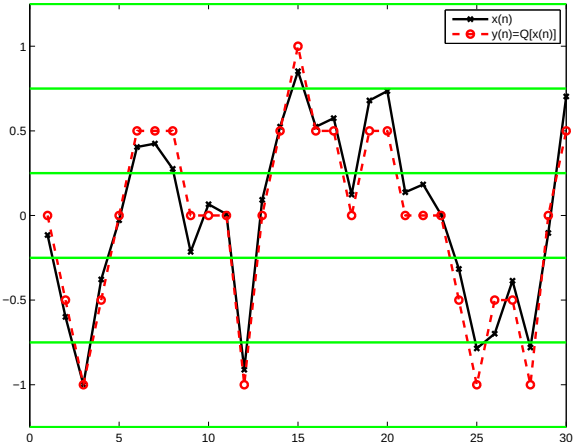


Go to **wooclap.com** with the code **PBTFUO**
Input interval:

Example 1



Go to **wooclap.com** with the code **PBTFUO**
Input interval: $(-A, A) = (-1.25, 1.25)$
Thresholds:

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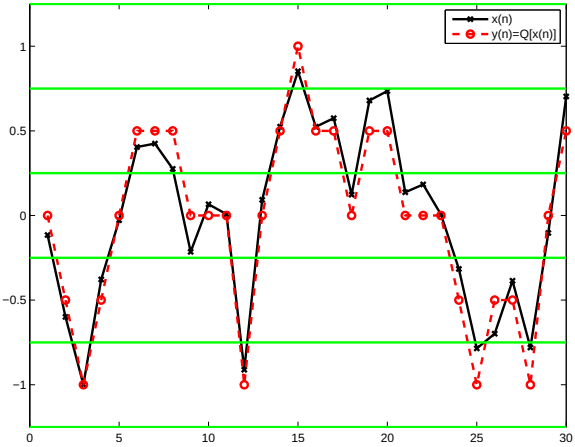

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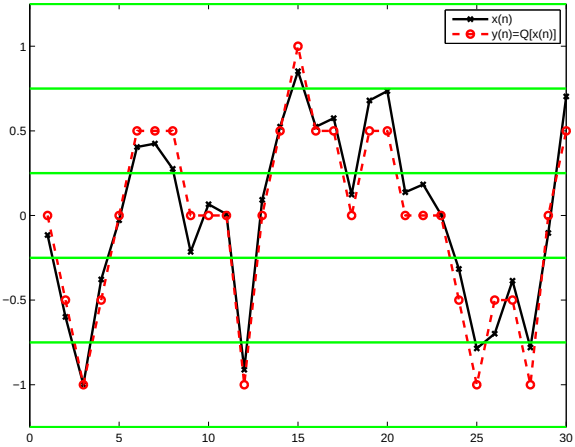

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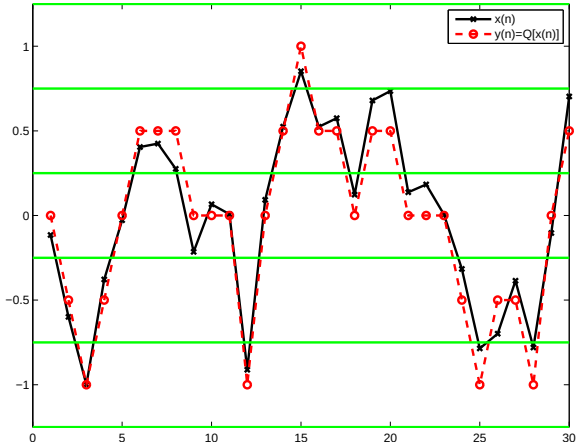

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Input interval: $(-A, A) = (-1.25, 1.25)$
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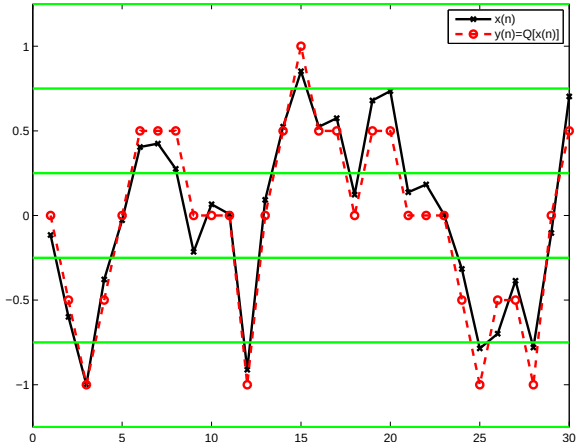

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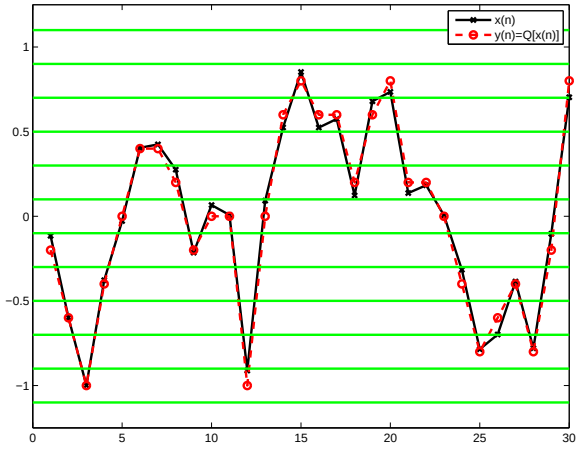
Levels: $-1, -0.5, 0, 0.5, 1$
Note that:

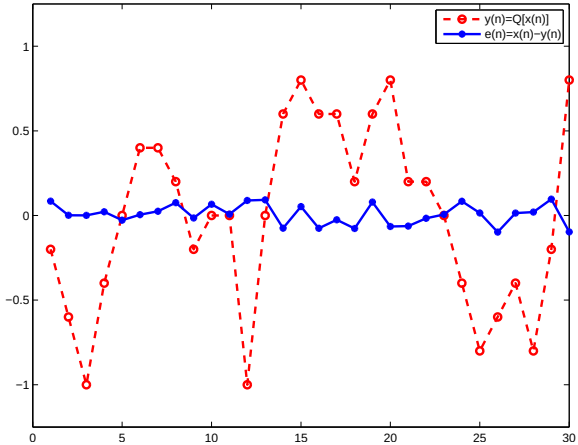
- All the cells have the same size $\Delta = \frac{2A}{L} = 2.5/5 = 0.5$
- Each level is the center of the corresponding cell

Example 2



Here we have again cells with identical size and levels at the center of the cell
The quantization cells are smaller $\Delta = 0.2$
The quantization error is typically smaller

Example 2



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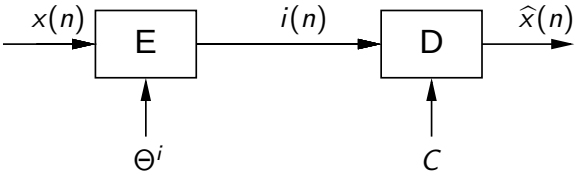
Quantization as coding / decoding



- The quantization process can be seen as the cascade of “encoding” and “decoding”
- The encoder produces the index of the quantization region, $i(n) \in \mathbb{Z}$
- The number $i(n)$ is represented with a suitable string of bits, e.g by using *lossless coding*
- The decoder associates to $i(n)$ to the corresponsing level (or code-word) $\widehat{x}^i(n)$
- In a compression system, often with the term quantization we just refer to the operation $x \to i$, while the evaluation of the quantization level, $i \to \widehat{x}^i$ is referred to as *Inverse Quantization*
  - The first is performed at the encoder, the second at the decoder

Rate of a scalar quantizer

- **Rate** R and **distortion** D are the performance criteria of a quantizer
- The rate of a scalar quantizer is the average number of bits needed to represent a sample, or rather, its quantization index: it amounts thus to the binarization of $i(n)$
- It is *always done via lossless coding*
- We will assume $R = \log_2 L$
- This is a good approximation of real-life coding rate, with two implicit assumption

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- This is a good approximation of real-life coding rate, with two implicit assumption
  - Quantized data have a uniform probability distribution (pessimistic)
  - The binary representation of $i(n)$ is the best (entropy coder)
- We will elaborate on this problem in the next lesson

Distortion of a scalar quantizer

- We use the squared error, which, on a single sample is:

$$d\left[x(n), Q[x(n)]\right] = |e(n)|^2 = |x(n) - Q[x(n)]|^2$$

- For a signal $x(\cdot)$ of duration N, we use the mean square error (MSE):

$$D = \frac{1}{N}\sum_{n=0}^{N-1} d[x(n), Q[x(n)]]$$

- For random signals[1], $D = \mathrm{E}\left\{|X(n) - Q(X(n))|^2\right\} = \mathrm{E}\left\{|E(n)|^2\right\}$
- In this case, distortion is the variance of the random process $E(n) = X(n) - Q(X(n))$, and is indicated as σ_{Q}^2

[1] Notice that we use lower-case letters for deterministic signals and upper-case letter for random variables/signals, while E is the expectation operator.

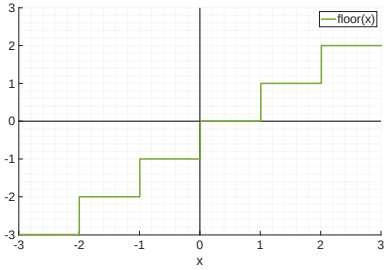
Mathematical reminders: integer mapping

- Let us remind the integer mapping functions commonly used in programming: `floor`, `ceil`, `round`, `fix`
- They are “staircase functions”
- They are widely used to implement quantizers

Mathematical Reminders: Floor Function

Operator: $\lfloor x \rfloor$

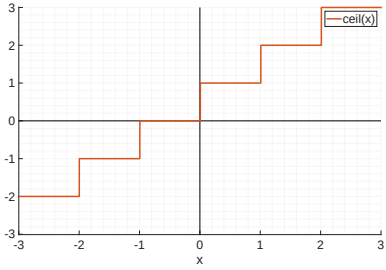
- **Definition**: The greatest integer less than or equal to x
- **Behavior**: Always rounds towards $-\infty$
- **Examples**:
  - $\lfloor 2.7 \rfloor = 2$
  - $\lfloor -2.7 \rfloor = -3$
- **Usage**: Common in basic uniform quantization models



Mathematical Reminders: Ceil Function

Operator: $\lceil x \rceil$

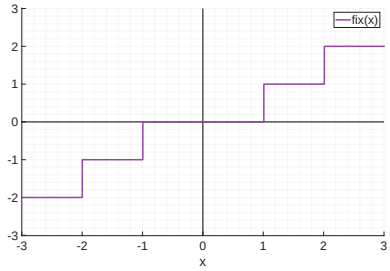
- **Definition**: The smallest integer greater than or equal to x
- **Behavior**: Always rounds towards $+\infty$
- **Examples**:
  - $\lceil 2.1 \rceil = 3$
  - $\lceil -2.1 \rceil = -2$
- **Usage**: Used for determining buffer sizes or resource allocation bounds



Mathematical Reminders: Fix Function

Operator: $\text{fix}(x)$

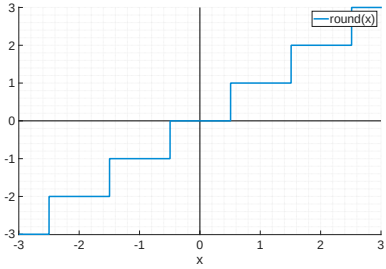
- **Definition**: Drops the fractional part of the number
- **Behavior**: Rounds towards zero (Symmetric)
  - Equivalent to $\lfloor x \rfloor$ for $x \geq 0$.
  - Equivalent to $\lceil x \rceil$ for $x < 0$.
- **Examples**:
  - $\text{fix}(2.7) = 2$
  - $\text{fix}(-2.7) = -2$
- **Note**: Standard behavior for integer casting in most programming languages



Mathematical Reminders: Round Function

Operator: round(x)

- **Definition**: Maps x to the nearest integer
- **Behavior**: Usually implemented as $\lfloor x + 0.5 \rfloor$
- **Examples**:
  - round$(2.51) = 3$
  - round$(2.49) = 2$
  - round$(-2.3) = -2$
- **Usage**: allow a straightforward implementation of *uniform quantization*



Outline

Uniform quantization

Uniform quantization is characterized by:

- The input range: $(0, A)$ (unsigned data) or $\left(-\frac{A}{2}, \frac{A}{2}\right)$ (signed data)
- The number of levels L

The input range is divided into L equal-sized cells. The cell size is $\Delta = \frac{A}{L}$. Each cell is represented by its mid-point, i.e. the quantization levels are the centers of the quantization cells.

$$\forall i,\ \Delta^i = \Delta = A/L$$

$$t^i = t^{i-1} + \Delta \text{ (red bars)}$$

$$\widehat{x}^i = \frac{t^i + t^{i-1}}{2} \text{ (green crosses)}$$

$$\Theta^i = \left(\hat{x}^i - \frac{\Delta}{2}, \hat{x}^i + \frac{\Delta}{2}\right)$$

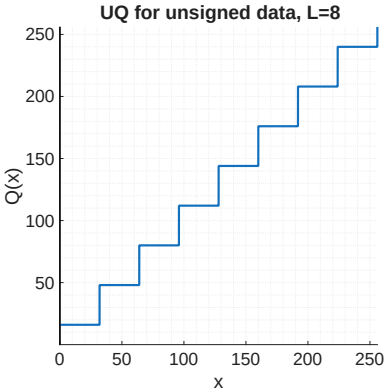
Uniform Quantization for unsigned data

For **unsigned data** (range $(0, A)$) the thresholds are $(0, \Delta, 2\Delta, \ldots, L\Delta)$, i.e., $t^i = (i-1)\Delta$.
We can thus define the behavior of the **encoder** (from x to the quantization index i), of the **decoder** (from i to the quantization level $\widehat{x}^i$) and of the full quantization chain:

$$i = \left\lceil \frac{x}{\Delta} \right\rceil \qquad \text{encoder}$$

$$\widehat{x}^i = i \cdot \Delta - \frac{\Delta}{2} \qquad \text{decoder}$$

$$Q(x) = \Delta \cdot \left\lceil \frac{x}{\Delta} \right\rceil - \frac{\Delta}{2}$$



Uniform quantization for signed data

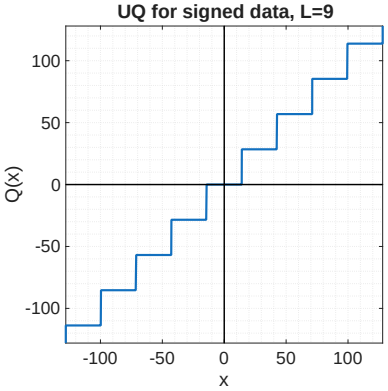
For **signed data** (range in $(-A/2, A/2)$) one typically wants zero to be a quantization level (**mid-tread** quantizer): in such a way, small fluctuations are quantized to zero. Therefore, we **always use an odd number of levels for UQ of signed data**. In this case, uniform quantization (UQ) amounts to **rounding**:

$$i = \text{round}\left(\frac{x}{\Delta}\right) \qquad \text{encoder}$$

$$\hat{x}^i = \Delta \cdot i \qquad \text{decoder}$$

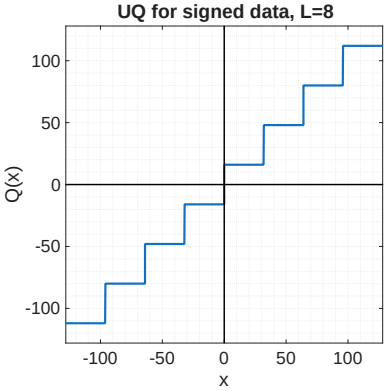
$$Q(x) = \Delta \cdot \text{round}\left(\frac{x}{\Delta}\right)$$

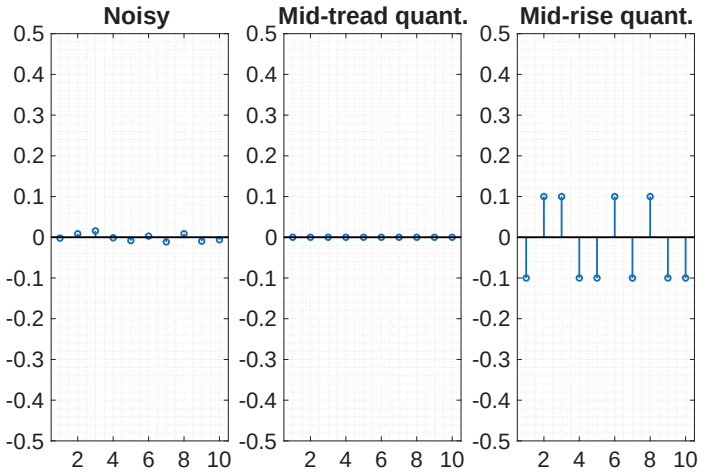
In other words, $Q(x)$ is **the closest multiple of** Δ. This quantizer is also referred to as **mid-tread** quantizer.



Uniform quantization for signed data

If one uses uniform quantization with an even number of levels for signed data, the central threshold $t^{\frac{N}{2}}$ is zero, see figure on the right.
This quantizer is also referred to as **mid-rise** quantizer.
Since zero is a threshold any very small value will be quantized as $\pm\Delta/2$ according to its sign.
This can be a problem is the signal has small fluctuations around zero due to additive noise: quantization enhances noise!





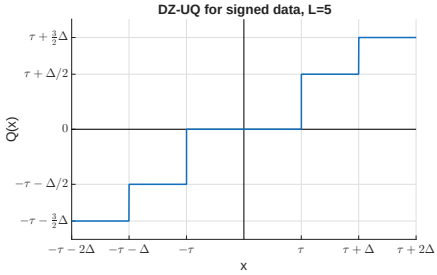
On the left, we show $x(n)$, a signal with small values that are **interpreted** as noise: uninformative, random fluctuations.
In the middle, we show $Q(x)$ when zero is a quantization level (**mid-tread** quantizer): the small values are forced to zero, making it "easier" to encode the signal.
On the right, we show $Q(x)$ when zero is a threshold (**mid-rise** quantizer): after quantization the noise is amplified instead of being removed!

Deadzone quantization

Pushing further this idea, we can design a quantizer where the central cell is larger than the others. This allows for removing more small-valued oscillations, while keeping a fine resolution on non-zero values. This is called **deadzone** quantization (DZQ) and is very commonly used in data compression.
Using the threshold value τ, the DZQ can be implemented as:

$$i = \begin{cases} \text{sign}(x) \cdot \left\lfloor \frac{|x| + \frac{\tau\Delta}{2}}{\Delta} \right\rfloor & \text{if } |x| \geq \tau \\ 0 & \text{if } |x| < \tau \end{cases} \qquad \text{encoder}$$

$$\hat{x} = \begin{cases} \text{sign}(i) \cdot \Delta \left(|i| + \frac{1-\tau}{2} \right) & \text{if } i \neq 0 \\ 0 & \text{if } i = 0 \end{cases} \qquad \text{decoder}$$



UQ: wrap-up

- UQ is intuitive: just split the input range into L intervals, and represent each interval with its middle.
- Implementation is easy:

$$Q(x) = \Delta \cdot \mathrm{round}\left(\frac{x}{\Delta}\right) \qquad \text{mid-tread}$$

$$Q(x) = \Delta \cdot \mathrm{floor}\left(\frac{x}{\Delta}\right) + \frac{\Delta}{2} \qquad \text{mid-rise}$$

- For a given range A and a given number of levels L, UQ has the smallest maximum error (optimal **minimax** quantizer)
  - The maximum error is half the size of the maximum quantization interval: any modification of the UQ increases the maximum quantization interval and thus the maximum error
- Last, but not least, UQ allows for an analytical formula of the RD (rate-distortion) curve. But before that...

Outline

Image quantization: example 1

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Rate 21 bpp PSNR 47.19 dB Compression ratio 1.143

Image quantization: example 1

Rate 18 bpp PSNR 42.38 dB Compression ratio 1.333

Image quantization: example 1

Rate 15 bpp PSNR 36.97 dB Compression ratio 1.600

Image quantization: example 1

Rate 12 bpp PSNR 31.40 dB Compression ratio 2.000

Image quantization: example 1

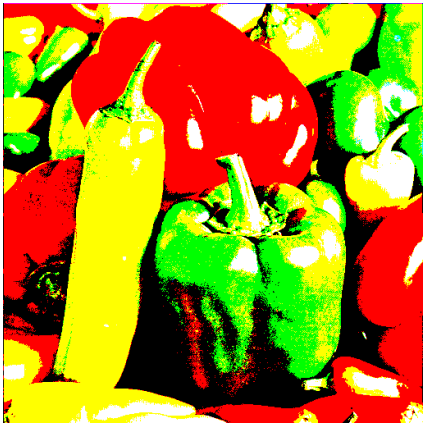
Rate 9 bpp PSNR 29.26 dB Compression ratio 2.667

Image quantization: example 1

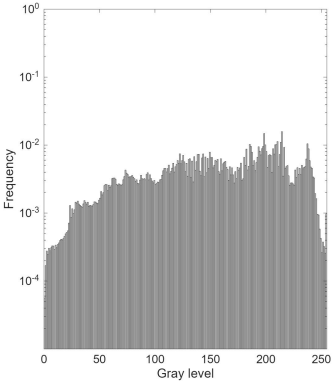
Rate 6 bpp PSNR 27.83 dB Compression ratio 4.000

Image quantization: example 1

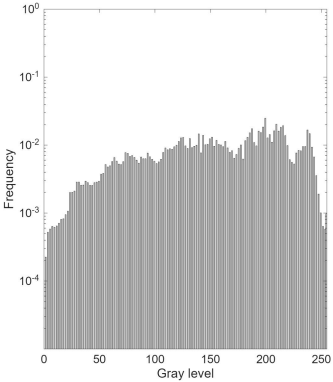
Rate 3 bpp PSNR 25.75 dB Compression ratio 8.000



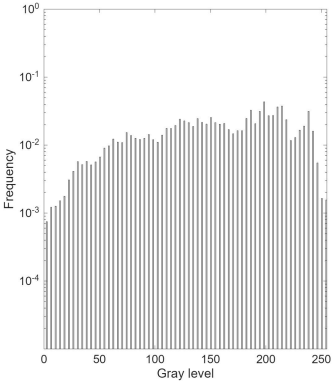
Scalar quantization: example 2



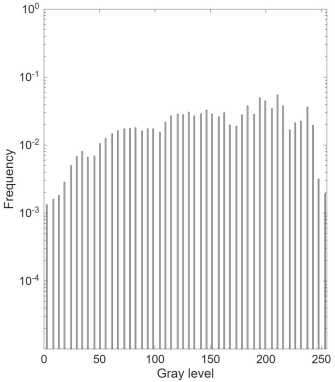
Scalar quantization: example 2



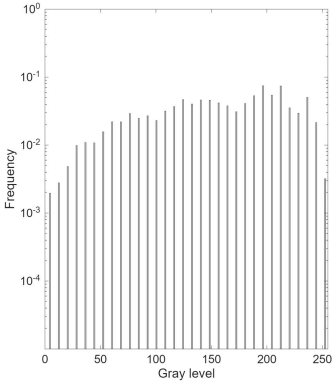
Scalar quantization: example 2



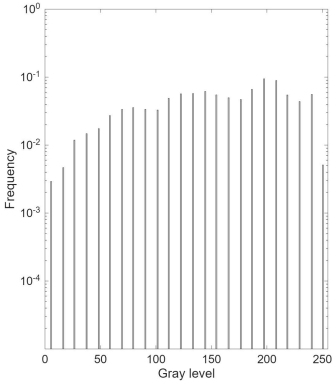
Scalar quantization: example 2



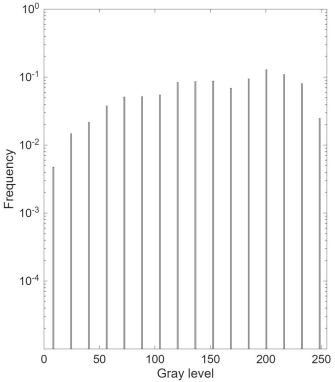
Scalar quantization: example 2



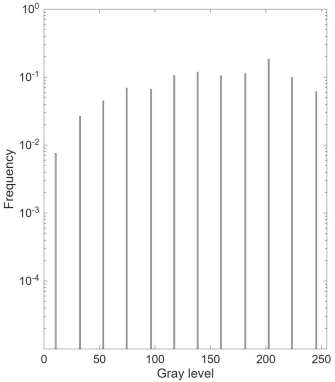
Scalar quantization: example 2



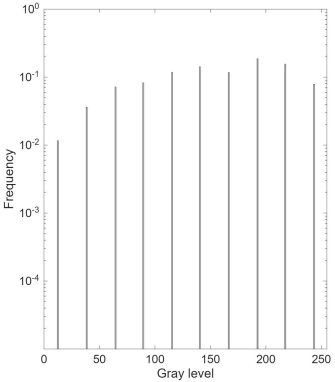
Scalar quantization: example 2



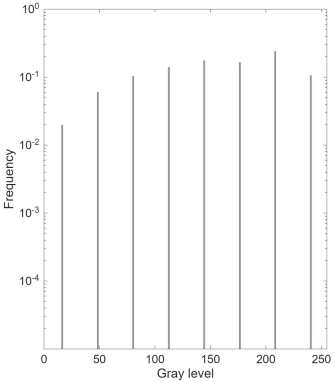
Scalar quantization: example 2



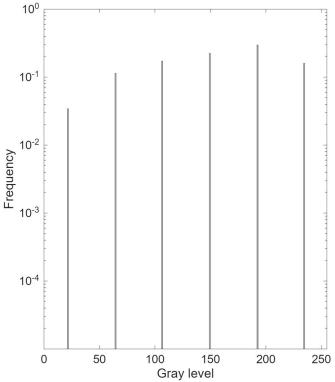
Scalar quantization: example 2



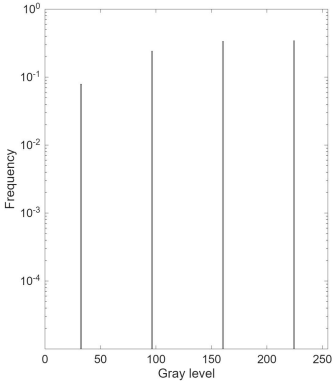
Scalar quantization: example 2



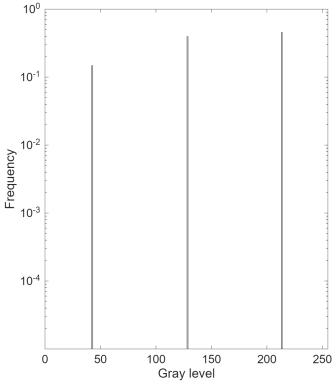
Scalar quantization: example 2



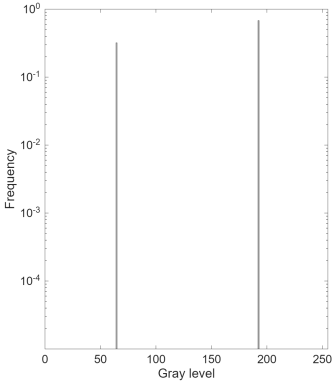
Scalar quantization: example 2



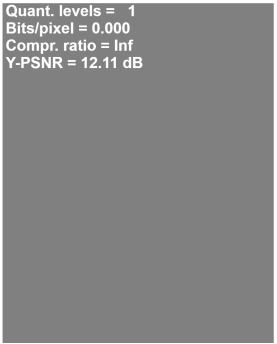
Scalar quantization: example 2

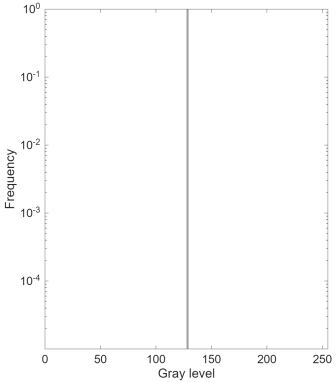


Scalar quantization: example 2

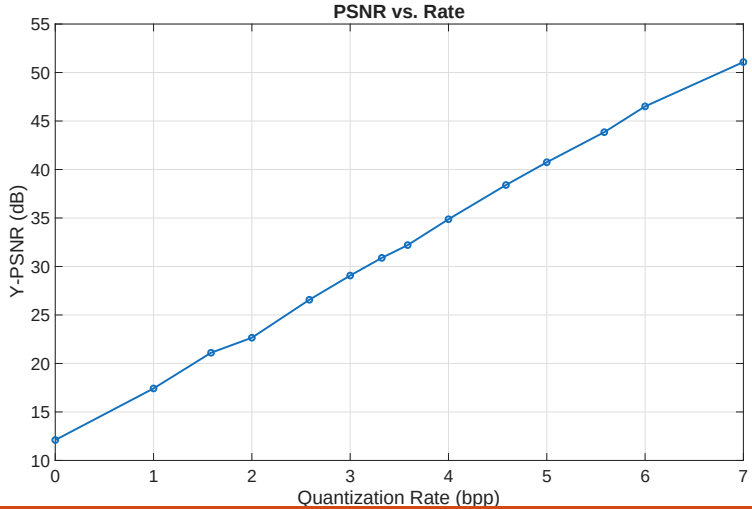


Scalar quantization: example 2

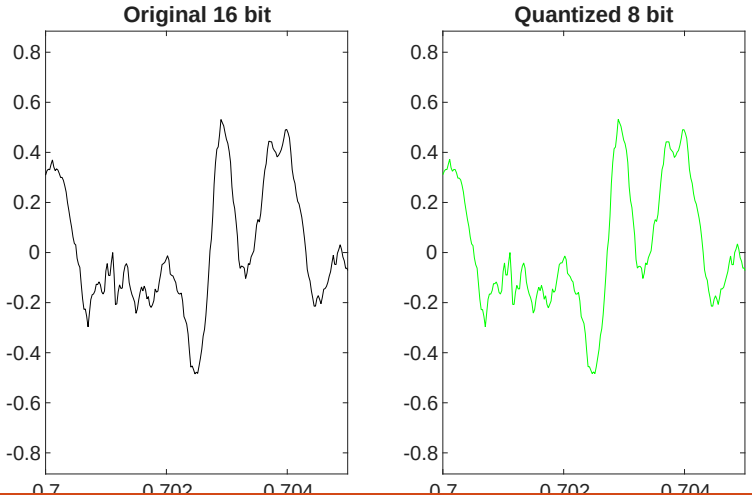




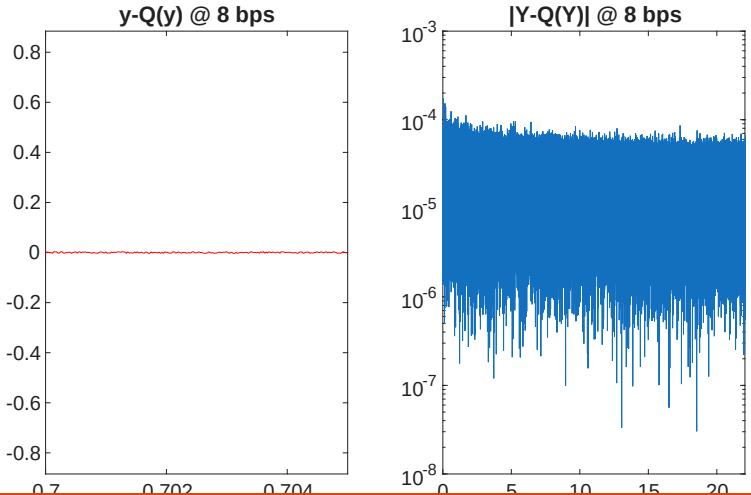
Scalar quantization: example 2



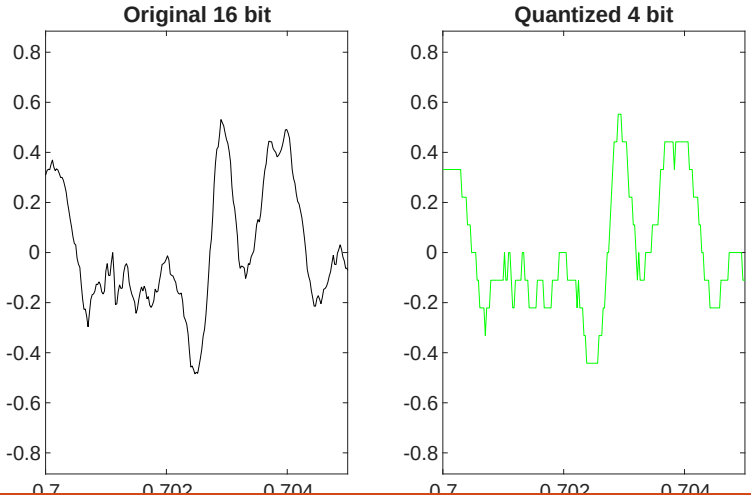
Audio quantization example



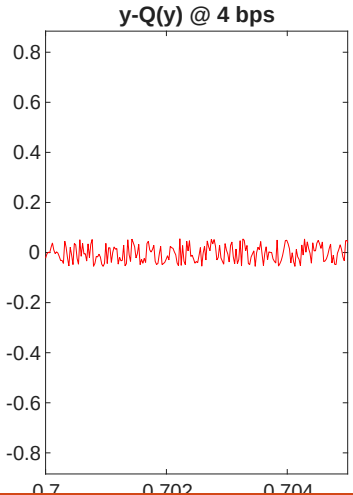
Audio quantization example

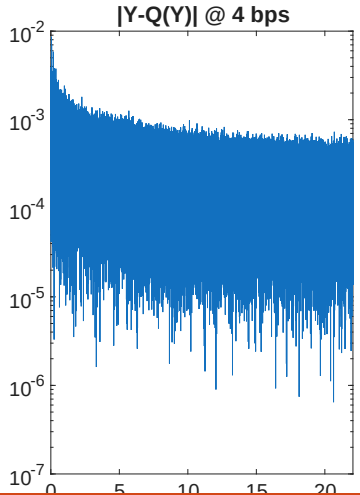


Audio quantization example

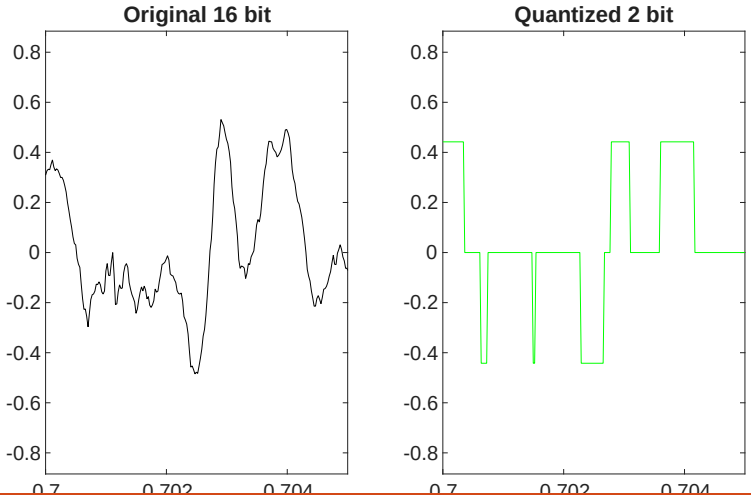


Audio quantization example

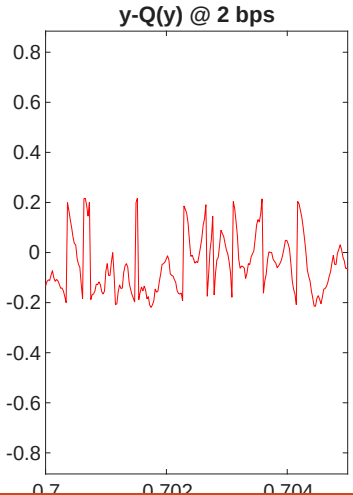


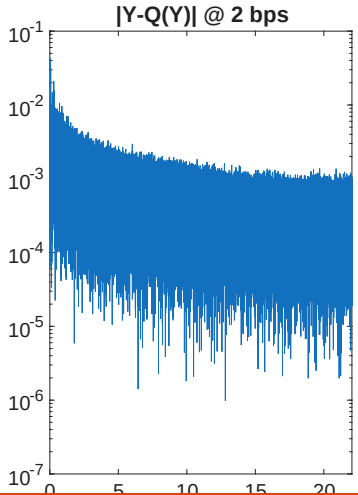


Audio quantization example



Audio quantization example





Audio Quantization: Subjective Comparison

Reference	Medium Rate	Low Rate	Very Low Rate
Original (16 bits)	Quantized (8 bits)	Quantized (4 bits)	Quantized (2 bits)

- **8-bit**: Notice the slight "hissing" or granular noise.
- **4-bit**: Observe the significant loss of fidelity and prominent quantization noise
- **2-bit**: The noise is stron; however the speech can still be understood, while the music is totally lost

Outline

Probability Theory Refreshers

1. Uniform Random Variable

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1. Uniform Random Variable

- A continuous random variable X is said to be **uniformly distributed** on the interval $[a, b]$ if its Probability Density Function (PDF) is defined as:

$$X \sim \mathcal{U}(a, b) \Leftrightarrow f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- Its variance is $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$

2. Expected Value of a Function of an R.V. (LOTUS)

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- Its variance is $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$

2. Expected Value of a Function of an R.V. (LOTUS)

- Let X be a continuous random variable with PDF $f_X(x)$. The expected value of a real-valued function $g(X)$ is given by:

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$$

- This theorem (Law of the Unconscious Statistician) allows us to calculate the mean of $g(X)$ without first finding its own PDF.

Uniform Quantization: RD curve

Let us apply these results to find the distortion of a uniform RV quantized with UQ.
Hypotheses : $X \sim \mathcal{U}\left(-\frac{A}{2}, \frac{A}{2}\right)$ and $Q(x)$ is a uniform quantizer with L levels.
Goal: Find $\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[(X - \widehat{X})^2\right]$ as a function of A, L

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We observe that $\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[g(X)\right]$, where $\forall u \in \mathbb{R}, g(u) = (u - Q(u))^2$.

$$\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[(X - \widehat{X})^2\right] = \int_{-\infty}^{+\infty} p_X(u)[u - Q(u)]^2 du$$

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$$
\begin{aligned}
\sigma_{\mathrm{Q}}^2 &= \mathrm{E}\left[(X - \widehat{X})^2\right] \\
&= \int_{-\infty}^{+\infty} p_X(u)[u - Q(u)]^2 du \\
\ldots &= \sum_{i=1}^{L} \int_{\Theta^i} \frac{1}{A}[u - \hat{x}^i)]^2 du \\
&= \frac{1}{A} \sum_{i=1}^{L} \int_{\hat{x}^i - \Delta/2}^{\hat{x}^i + \Delta/2} [u - \hat{x}^i)]^2 du \\
&= \frac{1}{A} \sum_{i=1}^{L} \int_{-\Delta/2}^{\Delta/2} t^2 dt \\
&= \frac{1}{A} L \frac{\Delta^3}{12} = \frac{\Delta^2}{12} = \frac{1}{12}\frac{A^2}{L^2}
\end{aligned}
$$

Uniform Quantization: RD curve

We can use the previous result to determine the **rate-distortion** curve of the UQ, i.e. the mathematical relationship between the rate R and the distortion $D = \sigma_{\mathrm{Q}}^2$.

$$D = \frac{\Delta^2}{12} = \frac{A^2}{12L^2} = \frac{A^2/12}{\cdot 2^{2R}} = \sigma_X^2 2^{-2R}$$

$$\mathrm{SNR} = 10 \log_{10} \frac{\mathrm{E}\{X^2\}}{D} = 10 \log_{10} \frac{\sigma_X^2}{\sigma_X^2 2^{-2R}}$$

$$= 10 \log_{10} 2^{2R} \approx 6R$$

High Resolution (HR) Uniform Quantization

- Hypothesis : $L \to +\infty$, X generic RV
- In HR, in any given Θ^i we approximate p_X as a constant (it is constant in the region, but the value can be different in another region)
- Therefore, the quantization noise in Θ^i is $\mathcal{U}\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$
- From the total probability law, $E \sim \mathcal{U}\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$
- Thus:

$$D = \frac{\Delta^2}{12} = \frac{A^2}{12} 2^{-2R}$$

High Resolution (HR) Uniform Quantization

$$\mathrm{SNR} = 10 \log_{10} \frac{\mathrm{E}\left\{X^2\right\}}{D} \qquad = 10 \log_{10} \frac{\sigma_X^2}{A^2/12} 2^{2R} \qquad \approx 6R - 10 \log_{10} \frac{\gamma^2}{3}$$

where $\gamma^2 = \frac{X_{\max}^2}{\sigma_X^2} = \frac{A^2/4}{\sigma_X^2}$ is the *load factor*, i.e. the ratio between the peak power and the average power.

$$D = \frac{A^2}{3} 2^{-2R} = \frac{\gamma^2}{3} \sigma_X^2 2^{-2R} = K_X \sigma_X^2 2^{-2R}$$

Outline

Optimal quantization

For a given probability density function $p_X(x)$, we want to find the quantizer minimizing distortion for a given rate
Problem equivalent to determine thresholds t^i and levels $\hat{x}^i$.
Solutions :

- Analytic Solution in *high resolution*: $D = c_X \sigma_x^2 2^{-2R}$
- If the high resolution hypothesis is not met, we can find a local optimum with the Lloyd-Max algorithm

Optimal quantization in high resolution

- The optimal quantizer is the one that, for a given PDF p_X of the input samples, provides the least distortion $D = \mathrm{E}\left[(X - Q(X))^2\right]$ for a given rate R
- It is possible to find the *rate-distortion* curve for the optimal quantizer in the **high-resolution hypothesis**:
  - $L \to \infty$
  - $\max_i \Delta_i \to 0$
  - $\forall i, \forall u \in \Theta^i, p_X(u) \approx P_i$
- Then it can be shown that the optimal quantizer has the following RD curve:

$$\sigma_Q^2 = c_X \sigma_X^2 2^{-2R} \qquad \text{with} \quad c_X = \frac{1}{12}\left[\int_{\mathbb{R}} p_U^{1/3}(t)dt\right]^3 \qquad \text{and} \quad U = \frac{X}{\sigma_X}$$

Shape Factor

In the RD curve of the optimal qunatizer

$$\sigma_Q^2 = c_X \sigma_X^2 2^{-2R}$$

the term c_X is called *shape factor*, since it only depends on the PDF *shape*, and not on its variance
The shape of X is obviously the same as of U

Shape factors for common PDFs:

	Uniform	Gaussian
c_X	1	$\frac{\sqrt{3}}{2}\pi \approx 2.72$

Non-HR Optimal quantization

- High resolution hypothesis is not always met
- On the contrary, often quantization must work at low or very low rate
- **There is no analytical solution for low resolution optimal quantization**
- On the other hand, it is possible to find **necessary conditions** for optimal quantizers independently from the resolution
- These conditions allows to define an algorithm attaining a local optimum point (Lloyd-Max algorithm)

Optimal quantization

Lloyd-Max algorithm

1. Choose an initial dictionary $\mathcal{C}^{(0)} = \{\hat{x}_0^i\}_{i=1,\dots L}$
2. Find the *best* thresholds for the given dictionary
3. Find the *best* dictionary for the given thresholds
4. Iterate 2-3 until a stop condition is met

Lloyd-Max algorithm: initial dictionary

- Uniform Initialization: $\widehat{x}^n = -A_0 + n\Delta \; \forall n = 1, \ldots, L$
- Random Initialization: code-words are initialized with the same distribution as data

Lloyd-Max algorithm: nearest neighbor rule

- Given $\{\hat{x}^i\}_{i=1,\dots L}$, we have to define, for each input, which is the best level to use as its representative
- The best choice is obviously the nearest neighbor:

$$k = \arg\min_n |x - \hat{x}^n| \Rightarrow Q(x) = \hat{x}^k$$

- Choosing the next neighbor is equivalent to set the threshold at the midpoint between consecutive quantization levels:

$$t^i = \frac{\hat{x}^i + \hat{x}^{i+1}}{2}, \quad i \in \{1, \dots, L-1\}$$

Lloyd-Max algorithm: centroid condition

Given the thresholds $\{t^0, \dots t^L\}$, the best levels are those minimizing the distortion:

$$\sigma_Q^2 = \sum_{i=1}^{L} \int_{t^{i-1}}^{t^i} (u - \widehat{x}^i)^2 p_X(u) du$$

To find the optimal levels, we compute the gradient of the distortion and set it to 0.

$$\frac{\partial \sigma_Q^2}{\partial \widehat{x}^i} = \frac{\partial}{\partial \widehat{x}^i} \int_{t^{i-1}}^{t^i} (u - \widehat{x}^i)^2 p_X(u) du = \int_{t^{i-1}}^{t^i} \frac{\partial}{\partial \widehat{x}^i} (u - \widehat{x}^i)^2 p_X(u) du$$

$$= \int_{t^{i-1}}^{t^i} 2(\widehat{x}^i - u) p_X(u) du = 2\widehat{x}^i \int_{t^{i-1}}^{t^i} p_X(u) du - 2 \int_{t^{i-1}}^{t^i} u p_X(u) du$$

$$\widehat{x}^i = \frac{\int_{t^{i-1}}^{t^i} u p_X(u) du}{\int_{t^{i-1}}^{t^i} p_X(u) du} = \frac{\int_{\Theta_i} u p_X(u) du}{\int_{\Theta_i} p_X(u) du} = \mathrm{E}\left[X | X \in \Theta_i\right]$$

This is the so-called centroid condition: the best representative of a quantization region is its centroid, i.e. the conditional average of X given $X \in \Theta^i$

Lloyd-Max algorithm: stop condition

- After the k-iteration of NN and the centroid rules, we can compute the new distortion value $\sigma^2_{\mathrm{Q},(k)}$
- It can be compared with the previous distortion $\sigma^2_{\mathrm{Q},(k-1)}$: the optimization steps assure that the new distortion can only be less than or equal to the old one
- We compute the relative distortion reduction: if it is smaller than a threshold, we stop the algorithm:

$$\frac{\sigma^2_{\mathrm{Q},(k-1)} - \sigma^2_{\mathrm{Q},(k)}}{\sigma^2_{\mathrm{Q},(k-1)}} < \varepsilon$$

- Another stop condition is upon a given number of iterations:

$$k = K$$

Convergence of Lloyd-Max algorithm

- Convergence to global optimum is not assured
- Final result depends on **initialization**
- However, at each step distortion cannot increase

Lloyd-Max algorithm for data

- In real-life problems, we have not signal PDFs, but rather a bunch of data
- We can modify Lloyd-Max algorithm as follows:
  1. Let $\mathcal{X} = \{u_1, u_2, \dots u_M\}$ be the set of data to quantize (or the training set if data is not known in advance)
  2. Initialize (k=0) with any dictionary (e.g. uniform): $\mathcal{C}^{(k)} = \{\widehat{x}_0^i\}_{i=1,\dots L}$
  3. Nearest neighbor rule:

$$W_k^i = \{u_m \in \mathcal{X} : \forall j \neq i \| u_m - \widehat{x}_k^i \| \leq \| u_m - \widehat{x}_k^j \| \}$$

  4. Centroid rule: $\widehat{x}_{k+1}^i = \frac{1}{|W_k^i|} \sum_{u_m \in W_k^i} u_m$
  5. Iterate to step 3 until convergence

Scalar quantization: summary

- $D(R)$, UQ of uniform RV: $D = \sigma_X^2 2^{-2R}$
- $D(R)$, UQ of non uniform RV in HR: $D = K_X \sigma_X^2 2^{-2R}$
- D(R), OQ in HR (any RV): $\sigma_Q^2 = c_X \sigma_X^2 2^{-2R}$
- c_X is the shape factor
  - $c_X = 1$ for uniform RVs
  - $c_X = \frac{\sqrt{3}}{2}\pi$ for Gaussian RVs
- At any resolution, the LMA produces a non-uniform quantizer which is locally optimal

Outline

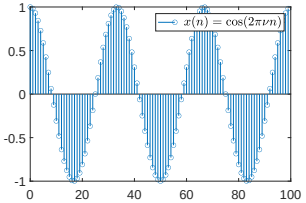
Predictive quantization

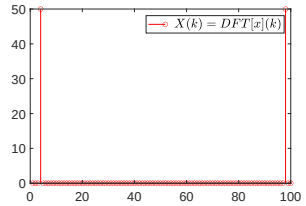
Principles

- Quantization alone is not effective for compression of **non-sparse data**
- Too simple underlying model: SQ neglects dependencies among samples
- Idea: exploit sample correlation by prediction
- Goal: reduction of the signal's variance by increasing sparsity

Sparse Signals: a preview

- **Definition:** A signal is sparse if most of its elements are zero or close to zero
- If a sample is close to zero, it can be "neglected" (quantized as zero) without introducing large distortion
- The variance of a sparse signal is low
- Natural signals are not sparse in their original form, but could be after some processing
- The main tools for signal sparsification are **prediction** and **linear transform**

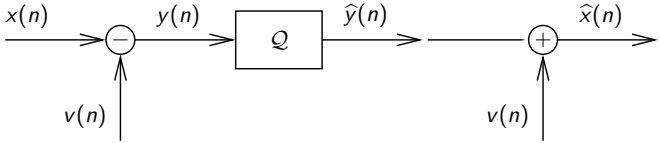




Coding scheme

Open loop scheme

- $x(n)$ is predicted from the past
- If the prediction is good, $v(n) \approx x(n)$



- How to provide $v(n)$?
- What is the gain?

Prediction gain

Prediction Error = Signal Error :

$$q(n) = y(n) - \hat{y}(n) = x(n) - v(n) - \hat{x}(n) + v(n) = \bar{q}(n)$$

Thus the goal of predictive quantization (PQ) is to minimize the distortion of y
The coding gain is:

$$\mathrm{SNR}_p = 10 \log_{10} \frac{\sigma_X^2}{D} = 10 \log_{10} \frac{\sigma_X^2}{\sigma_Y^2} + 10 \log_{10} \frac{\sigma_Y^2}{D} = G_P + G_Q$$

Prediction is effective if and only if the prediction error has a smaller variance than the original signal

Example

Go to **wooclap.com** with the code **PBTFUO**

$$X(n) \sim \mathcal{N}(0, \sigma^2) \qquad \mathrm{E}\left[X(n)X(m)\right] = \sigma^2 \rho^{|n-m|}$$

$$V(n) = X(n-1) \qquad \rho : G_P > 0 \ ?$$

Example

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$$V(n) = X(n-1) \qquad \rho : G_P > 0 \ ?$$

$$Y(n) = X(n) - X(n-1) \text{ Zero mean Gaussian RV}$$

$$\sigma_Y^2 = \mathrm{E}\left[(X(n) - X(n-1))^2\right] = 2\sigma^2 - 2\sigma^2\rho$$

$$G_P = 10 \log_{10} \frac{\sigma_X^2}{\sigma_Y^2} = 10 \log_{10} \frac{\sigma^2}{2(1-\rho)\sigma^2}$$

$$G_P > 0 \Leftrightarrow \rho > \frac{1}{2}$$

Predictors

- Linear Predictors are simple and moreover optimal for Gaussian RV

$$v(n) = -\sum_{i=1}^{P} a_i x_{n-i} \qquad \text{Filter with } P \text{ parameters}$$

$$y(n) = x(n) - v(n) = \sum_{i=0}^{P} a_i x_{n-i} \qquad \text{Prediction error}$$

- with $a_0 = 1$.
- y is the result of filtering x with

$$A(z) = 1 + a_1 z^{-1} + \ldots + a_P z^{-P}$$

- Optimal filter: minimization of σ_Y^2

Predictor selection

Problem:
Find the vector $\underline{a}$ minimizing

$$\sigma_Y^2 = \mathrm{E}\left\{Y^2(n)\right\} = \mathrm{E}\left\{\left[X(n) + \sum_{i=1}^{P} a_i X(n-i)\right]^2\right\}$$

Predictor selection

$$\sigma_Y^2 = \mathrm{E}\left\{X^2(n)\right\} + 2\sum_{i=1}^{P} a_i \mathrm{E}\{X(n)X(n-i)\} + \sum_{i=1}^{P}\sum_{j=1}^{P} a_i a_j \mathrm{E}\{X(n-i)X(n-j)\}$$
$$= \sigma_X^2 + 2\underline{r}^t\underline{a} + \underline{a}^t \mathrm{R}_X \underline{a}$$

with:

$$\underline{r} = [r_X(1) \dots r_X(P)] \quad \mathrm{R}_X = \begin{bmatrix} r_X(0) & r_X(1) & r_X(2) & \dots & r_X(P-1) \\ r_X(1) & r_X(0) & r_X(1) & \dots & r_X(P-2) \\ r_X(2) & r_X(1) & r_X(0) & \dots & r_X(P-3) \\ \dots & \dots & \dots & \dots & \dots \\ r_X(P-2) & r_X(P-3) & r_X(P-4) & \dots & r_X(1) \\ r_X(P-1) & r_X(P-2) & r_X(P-3) & \dots & r_X(0) \end{bmatrix}$$

$$r_X(k) = \mathrm{E}\{X(n)X(n-k)\}$$

Predictor selection

Minimization of Y variance:

$$\frac{\partial \sigma_Y^2}{\partial \underline{a}} = 2\underline{r} + 2\mathrm{R}_X \underline{a} = 0$$

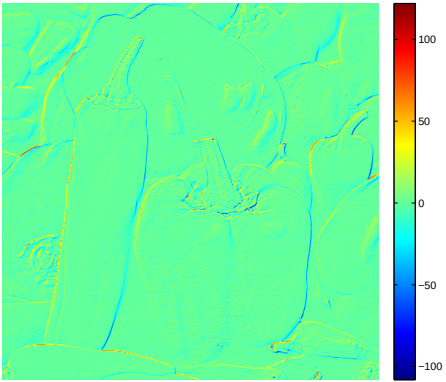
Thus:

$$\underline{a}^{\mathrm{opt}} = -\mathrm{R}_X^{-1}\underline{r} \qquad \sigma_Y^2 = \sigma_X^2 + \underline{r}^t \underline{a}^{\mathrm{opt}}$$

Autocorrelation r_X can b estimated as

$$\widehat{r}_X(k) = \frac{1}{N}\sum_{n=0}^{N-1-k} X(n)X(n+k)$$

Predictive quantization: example



For the "peppers" image (grayscale), we compute the prediction error using a second order linear predictor:

$$\hat{f}_{n,m} = af_{n-\mathbf{1},m} + bf_{n,m-\mathbf{1}}$$

The predictor of $f_{n,m}$ is a weighted average of the top and the left neighbors.

a	b	σ^2_Y
0	0	2902.7
1/2	1/2	78.7
0.449	0.546	78.4

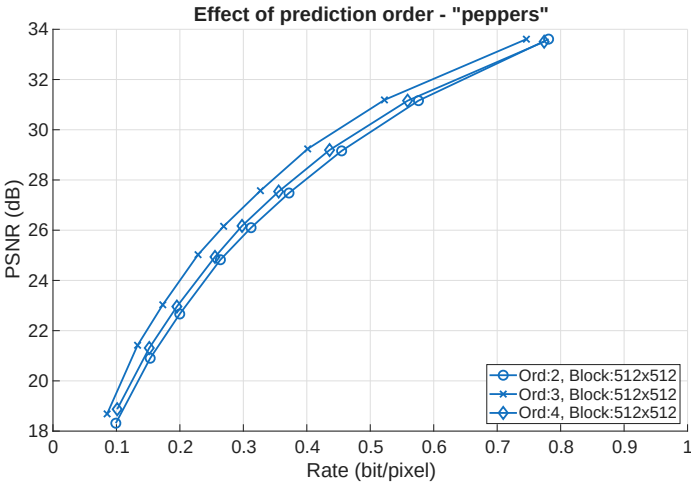
The first row correspond to no prediction: thus $\sigma^2_X = \mathbf{2902.7}$
The second row show the distortion of the simplest predictor: the average of the two neighbors. The distorion is reduced by a factor greater than 37
Using the optimal predictor, computed with the formulas shown in the previous slide, only slightly improves the performance

Impact of Filter Order on Prediction Gain

In global DPCM, the predictor $v(n)$ uses a linear combination of N causal neighbors.

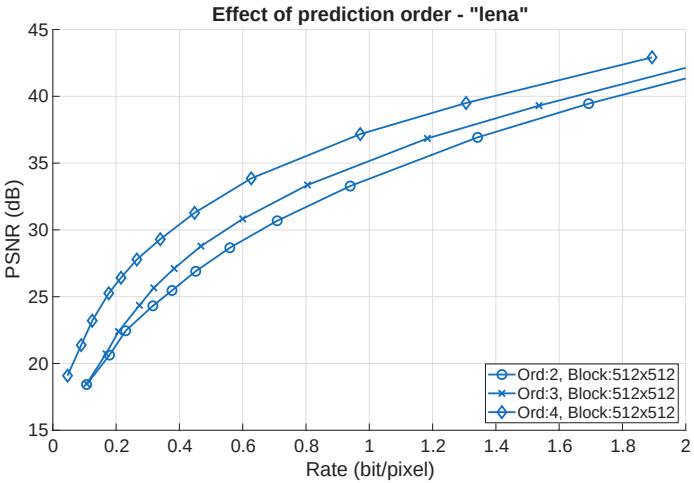
- **Order Selection:** Increasing the order from 2 to 4 typically improves the **Prediction Gain** ($G_p = \sigma_X^2/\sigma_Y^2$), leading to a more "whitened" residual.
- **Screening Effect:** The benefit of including "distant" pixels diminishes rapidly. In typical image statistics, the immediate neighbors capture most of the mutual information. Distant pixels are often redundant as their correlation with the target is already "screened" by the closer neighbors.
- **Complexity vs. Performance:** Higher orders increase computational overhead with negligible PSNR gains for most natural textures.

Rate-Distortion Analysis: Filter Order



Homogeneous image with some background noise

Rate-Distortion Analysis: Filter Order



Clean image with different areas

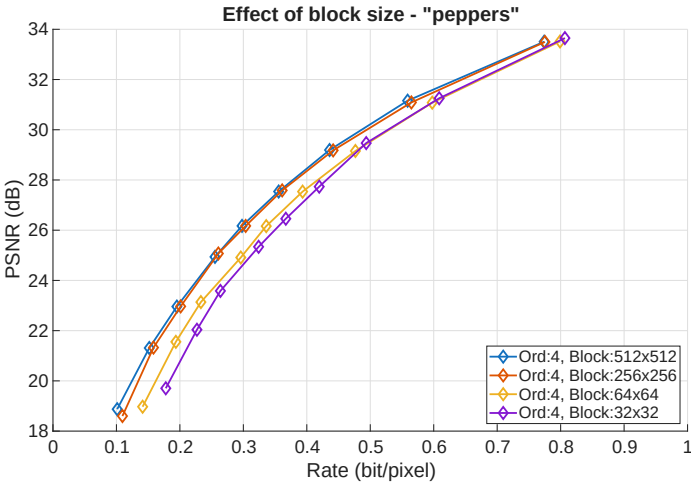
Local Adaptation: The Block-wise Approach

Images are **non-stationary** signals; a single global filter is suboptimal for varying textures (e.g., edges vs. smooth areas). **The Strategy:** Divide the image into $M \times M$ blocks and

compute the **locally optimal** filter for each.
The Side Information Trade-off Total Rate (R_{total}) is now increased by $\frac{N \cdot B}{M^2}$ Where N is the filter order and B is the bits per coefficient (e.g., 16).

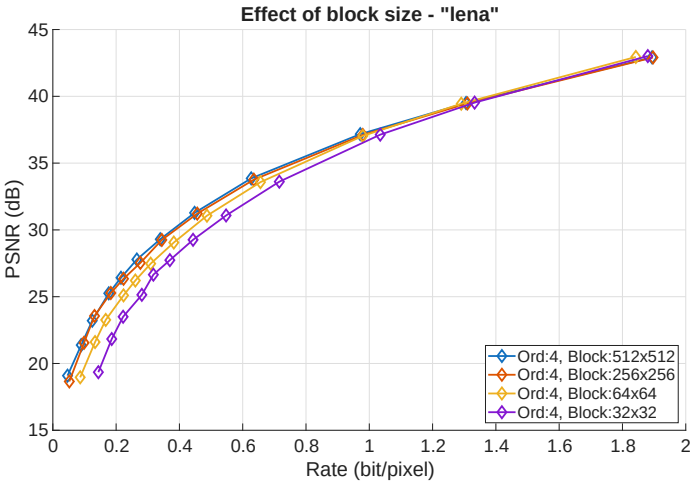
- **Small Blocks:** Excellent tracking of local statistics, but massive **overhead** from filter coefficients.
- **Large Blocks:** Minimal overhead, but the predictor fails to adapt to local features.

Rate-Distortion Analysis: Block size



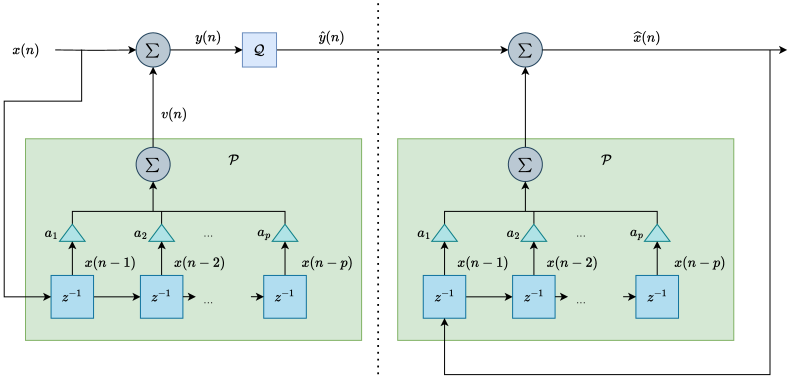
Homogeneous image with some background noise

Rate-Distortion Analysis: Block size

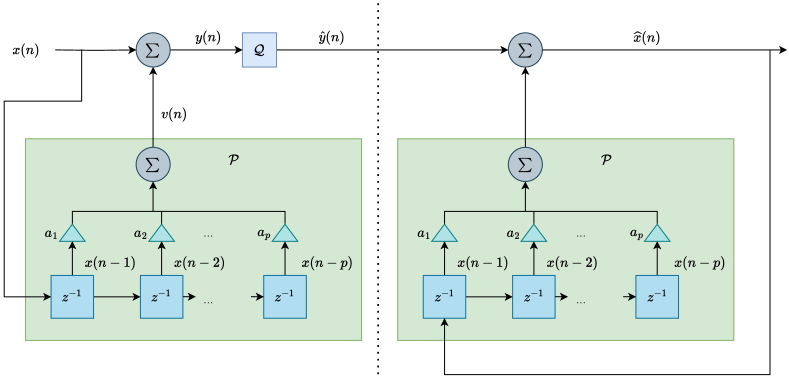


Clean image with different areas

Coding/decoding scheme



Coding/decoding scheme



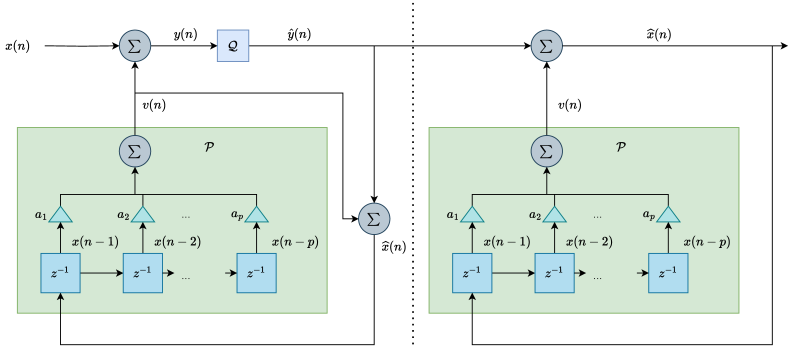
WRONG! The predictor operator is the same at the encoder and decoder but the input is different: $x \neq \hat{x}$! This causes **drift**.

Predictive Quantization Drift Example

- Simple predictive scheme: prediction is the previous value
- 3-bit quantizer (7 levels): [-9, -6, -3, 0, 3, 6, 9]
- Encoder uses non-quantized values for prediction
- Decoder uses quantized (reconstructed) values for prediction

Step	Original Signal	Encoder			Decoder		
		Prediction	Error	Quantized Error	Prediction	Received Error	Reconstructed Value
1	10	-	-	9	-	9	9
2	11	10	1	0	9	0	9
3	12	11	1	0	9	0	9
4	13	12	1	0	9	0	9
5	14	13	1	0	9	0	9
6	18	14	4	3	9	3	12
7	21	18	3	3	12	3	15
8	18	21	-3	-3	15	-3	12

Coding/decoding scheme



This is the correct scheme: the predictor is fed with the same data (quantized past samples) both at the encoder and the decoder.

Predictive Quantization Drift Example

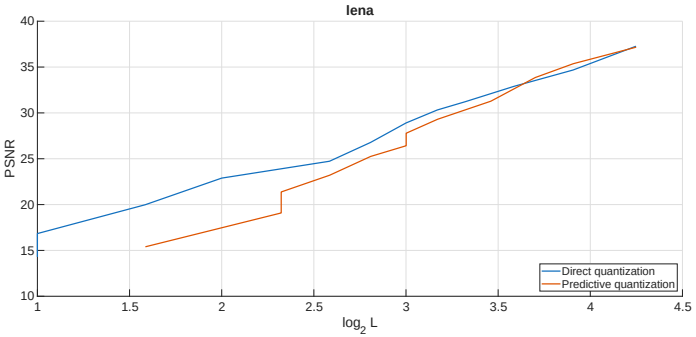
- Simple predictive scheme: prediction is the previous value
- 3-bit quantizer (7 levels): [-9, -6, -3, 0, 3, 6, 9]
- Encoder uses quantized values for prediction
- Decoder uses quantized (reconstructed) values for prediction

Step	Original Signal	Encoder			Decoder		
		Prediction	Error	Quantized Error	Prediction	Received Error	Reconstructed Value
1	10	-	-	9	-	9	9
2	11	9	2	3	9	3	12
3	12	12	0	0	12	0	12
4	13	12	1	0	12	0	12
5	14	12	2	3	12	3	15
6	18	15	3	3	15	3	18
7	21	18	3	3	18	3	21
8	18	21	-3	-3	21	-3	18

Coding scheme

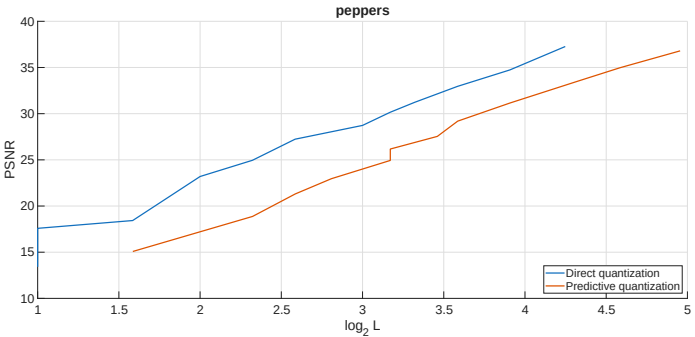
- The encoder computes the prediction error $Y = X - V$
- The predictor V must be computed from data available at the decoder as well
- Thus the encoder must compute the reconstructed value $\widehat{X}$
- This value is sent into a buffer and used to compute the prediction of future values

Direct vs. Predictive quantization



We show here the PSNR of the quantized image vs. the $log_2 L$, which is our first rate estimation. The result seems very deceiving: apart from a short range of high rates, predictive quantization does not bring the expected gain: how is this possible?
The main explanation is that a practical PQ scheme must use the quantized data to produce a predictor, which is not what we considered in our theoretical analysis.

Direct vs. Predictive quantization

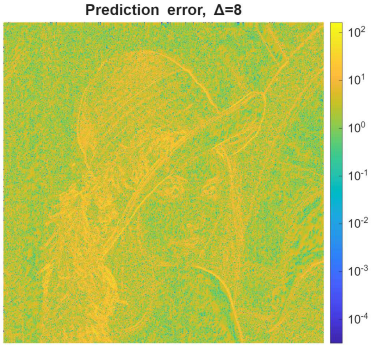


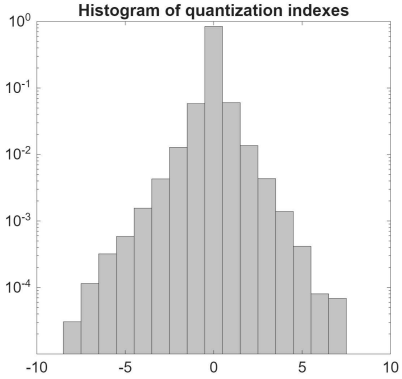
For the **peppers** image things are even worse.

- At low bitrates, the predictor is computed on heavily quantized data
- At high bitrates, the quantizer is obliged to reconstruct the additive noise affecting the image

However,the main point is that **the estimation of the rate with the logarithm is too much pessismistic**

PQ prediction error





PQ prediction error

Let us look more closely to the quantized prediction error. In this example, $L = 19$, so the estimated rate is $\log_2 19 \approx 4.25$ bits per sample.
However, we observe that most of the quantization indexes ($\approx 84\%$) are zero. We could use a **variable lenght code** that uses a short codeword for the index 0 and possibly longer codewords for indexes outside $(-3, 3)$, which account for less than 1% of the total.
Encoding the quantizations indexes with non-trivial encoding strategy (called **entropy coding**, as we will see later on) changes dramatically the results

Direct vs. Predictive quantization

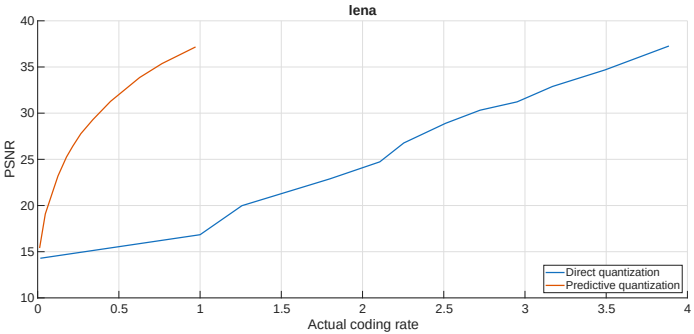
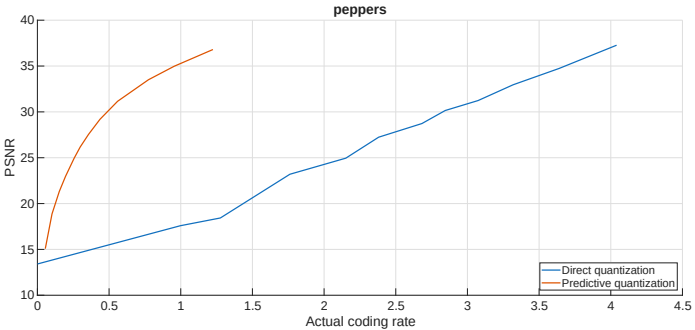


Image: **lena**
Using PQ+entropy coding is much better that UQ+entropy coding: we achieve an impressive $\Delta\mathbf{PSNR} = 20$ **dB** at 1 bpp or a -86 % **rate reduction** at 30dB

Direct vs. Predictive quantization



Results for **peppers** are similar

Quantization: wrap-up

- UQ: $D = K_X \sigma_X^2 2^{-2R}$ for uniform RV or in high res
- Optimal SQ: similar behavior
- Both UQ and optimal SQ alone are catastrophic is applied directly on images or sound: the quality drops and the rate is not reduced much
- **Predictive quantization** can improve dramatically performance but:
  1. Never forget to compute the prediction on **quantized data**, otherwise the decoder will drift
  2. The input signal **must be correlated enough**
  3. **We must use the best encoding strategy to achieve good performance**

The best encoding strategy is **entropy coding**, the subject of the next lesson